Plant Structure Pipeline - Tunable Parameters

All settings live in the `PipelineConfig` class at the top of `copy_of_New_optimized_plant_structure.py`. Change a default there, or, where a command-line flag is listed, pass it when running the script, e.g. `python copy_of_New_optimized_plant_structure.py --voxel_size 0.003 --leaf_cluster_eps 0.02`. Distances are in metres. Parameters with '-' in the flag column can only be changed in the code.

Step 1 - Cleaning, downsampling and normals

Removes noise from the raw scan, thins it to an even grid and estimates a surface normal for every point.

Parameter	Default	Command-line flag	What it does	Effect of changing it
`sor_neighbors`	20	`--sor_neighbors`	Statistical outlier removal: each point's mean distance to its k nearest neighbours is computed.	Higher = smoother, more stable noise estimate; lower = reacts to very local noise.
`sor_std_ratio`	2.0	`--sor_std_ratio`	A point is kept if its mean neighbour distance is below global mean + ratio x std.	Lower = removes more points (also thin leaf edges); higher = keeps more noise.
`ror_radius`	0.012 m	`--ror_radius`	Radius outlier removal: size of the search ball around each point (1.2 cm).	Larger = more lenient; smaller = stricter, may strip sparse leaf tips.
`ror_min_neighbors`	8	`--ror_min_neighbors`	A point with fewer neighbours than this inside ror_radius is deleted as noise.	Higher = stricter cleaning; lower = keeps isolated points.
`voxel_size`	0.004 m	`--voxel_size`	Grid cell size for downsampling: all points in a 4 mm cube become one point. Most later steps run on this cloud.	Larger = faster but coarser (width/area lose detail); smaller = slower, more detail and more noise.
`normal_radius`	0.012 m	-	Fixed neighbourhood radius for normal estimation. Only used when use_adaptive_normal_radius is False.	Larger = smoother normals; smaller = sharper but noisier.
`normal_radius_factor`	3.0	-	Adaptive normal radius = factor x voxel_size (12 mm by default).	Same effect as normal_radius, but scales with voxel_size.
`use_adaptive_normal_radius`	True	-	Use voxel_size x normal_radius_factor instead of the fixed normal_radius.	Keep True so the normals follow any voxel_size change.
`normal_max_nn`	40	-	Maximum number of neighbours used for one normal, even if more fall inside the radius.	Higher = smoother normals in dense areas; lower = faster, keeps detail.

Step 2 - Stem / leaf segmentation: features and stem axis

Each point is described with local PCA shape features (linearity, planarity, curvature) and its distance to the estimated stem axis.

Parameter	Default	Command-line flag	What it does	Effect of changing it
`feature_k`	48	`--feature_k`	Number of neighbours for PCA features. Only used if feature_scales is empty.	Higher = smoother features; lower = more local detail.
`feature_scales`	(20, 40, 80)	-	Multi-scale PCA: features are computed with 20, 40 and 80 neighbours and averaged.	Add small k for thin stems / fine detail, large k for robustness to noise.
`stem_radial_max`	0.0345 m	-	Nominal maximum stem radius (3.45 cm). Points farther from the stem axis are unlikely to be stem.	Raise for thick stems; lower if leaf bases are classified as stem.
`stem_radial_cap_factor`	1.5	-	Adaptive stem radius may not exceed 1.5 x stem_radial_max (about 5.2 cm).	Upper safety limit for the adaptive radius.
`stem_radial_floor_factor`	0.85	-	Adaptive stem radius may not drop below 0.85 x stem_radial_max (about 2.9 cm).	Lower safety limit for the adaptive radius.
`use_adaptive_stem_radius`	True	-	Estimate the stem radius from this plant instead of always using stem_radial_max.	Turn off to force a fixed radius.
`adaptive_stem_radius_sigma`	1.0	-	Adaptive radius = median + sigma x std of the distances of the innermost 30% of lower-plant points.	Higher = wider stem radius (more points counted as stem).
`stem_verticality_min`	0.55	-	Stem surfaces must be more vertical than horizontal. Filters flat leaf blades.	Higher = stricter stem; lower = tilted stem parts accepted.
`stem_linearity_min`	0.35	-	Neighbourhood must be at least this line-like to count as stem (soft gate).	Higher = only clearly cylindrical parts are stem.
`stem_curvature_max`	0.25	-	Stem neighbourhoods must have curvature below this. High curvature = junction or leaf.	Lower = stricter stem, junctions go to leaves.
`leaf_planarity_min`	0.18	`--leaf_planarity_min`	Minimum planarity for a point to be called leaf-like.	Higher = only flat blade points are leaf; lower = curled leaf parts accepted.
`root_xy_cell`	0.01 m	-	XY grid cell (1 cm) used to find the stem axis from the point density.	Smaller = more precise axis on clean data; larger = more robust on sparse data.
`root_density_z_frac`	0.55	-	Only the lowest 55% of the plant height is used to find the stem axis.	Lower if leaves near the bottom are dense and pull the axis sideways.

Step 2 - Stem score

Stem score = weighted sum of four cues, each scaled to 0..1. Weights are normalised by their sum, so they need not add up to 1.

Parameter	Default	Command-line flag	What it does	Effect of changing it
`stem_score_weight_radial`	0.40	-	Weight of 'close to the stem axis'. The most important cue.	Raise if the stem is straight and vertical.
`stem_score_weight_linearity`	0.30	-	Weight of 'neighbourhood is line-like' (stems are thin cylinders).	Raise for thin, clean stems.
`stem_score_weight_curvature`	0.20	-	Weight of 'low curvature' (smooth surface, not a junction).	Raise to keep junctions out of the stem.
`stem_score_weight_verticality`	0.10	-	Weight of 'surface is vertical'.	Raise for upright plants; lower for leaning stems.
`stem_score_threshold`	0.58	`--stem_score_threshold`	Points with stem score >= threshold are the initial stem candidates.	Higher = smaller, cleaner stem; lower = more stem (may include leaf bases).

Step 2 - Region growing and clean-up

Starts from very confident stem seed points and grows into neighbours that look similar. Then removes tiny leaf blobs.

Parameter	Default	Command-line flag	What it does	Effect of changing it
`use_region_growing`	True	`--no_region_growing (turns off)`	Turn region growing on or off.	Off = only the score threshold decides.
`stem_seed_linearity_min`	0.60	-	Seed points must be at least this line-like.	Lower if no seeds are found on short stems.
`stem_seed_curvature_max`	0.12	-	Seed points must have curvature below this (very smooth).	Higher = more seeds.
`stem_seed_radial_frac`	0.55	-	Seed points must lie within 55% of the stem radius (stem core).	Higher = more seeds near the stem edge.
`region_grow_k`	16	-	Number of neighbours checked around each grown point.	Higher = grows faster and further.
`region_grow_normal_sim_min`	0.75	-	A neighbour joins only if its normal is within ~41 deg (dot product >= 0.75).	Higher = stricter, stops at bends; lower = may leak into leaves.
`region_grow_curvature_sim`	0.08	-	Maximum curvature difference between a point and its neighbour.	Higher = more leaking.
`region_grow_linearity_sim`	0.15	-	Maximum linearity difference between a point and its neighbour.	Higher = more leaking.
`region_grow_radial_sim`	0.012 m	-	Maximum difference in distance from the stem axis (1.2 cm).	Higher = more leaking.
`min_leaf_component_size`	50	`--min_leaf_component_size`	Connected leaf blobs with fewer points are dropped as noise.	Higher = cleaner but small leaves can disappear.
`leaf_component_graph_k`	12	-	k of the k-NN graph used to find connected leaf blobs.	Higher = blobs connect more easily.

Step 3 - Plant k-NN graph

A neighbourhood graph over the whole downsampled plant.

Parameter	Default	Command-line flag	What it does	Effect of changing it
`graph_k`	12	`--graph_k`	Each point is linked to its k nearest neighbours.	Higher = denser graph, slower.
`graph_max_edge`	0.07 m	-	Edges longer than 7 cm are removed (also used when finding leaf blobs).	Lower = gaps split parts apart; higher = distant parts get linked.

Step 4 - Stem centerline (Laplacian contraction)

Shrinks the stem points onto their centre line, then slices by height to get an ordered centerline.

Parameter	Default	Command-line flag	What it does	Effect of changing it
`laplacian_k`	16	-	Neighbours used for the local mean in each contraction step.	Higher = smoother, faster contraction.
`laplacian_iterations`	35	`--laplacian_iterations`	Number of contraction iterations.	More = thinner skeleton; too many can shorten the stem.
`laplacian_contract`	0.55	-	How far each point moves toward its neighbours' mean per iteration (0..1).	Higher = faster contraction.
`laplacian_volume`	0.12	-	Pull back toward the original position each iteration so the stem does not collapse.	Higher = keeps original shape more.
`stem_z_step`	0.008 m	-	Height slice (8 mm); each slice's mean is one centerline point.	Smaller = more centerline points, more detail and noise.
`stem_smooth_window`	5	-	Moving-average window (points) for smoothing the centerline in X and Y.	Higher = straighter centerline.

Step 5 - Leaf clusters (individual leaves)

DBSCAN groups leaf points into leaves; large clusters that span a wide angle around the stem are split with KMeans.

Parameter	Default	Command-line flag	What it does	Effect of changing it
`leaf_cluster_eps`	0.028 m	`--leaf_cluster_eps`	DBSCAN radius (2.8 cm) when adaptive DBSCAN is off. In adaptive mode eps is at least half of this.	Larger = touching leaves merge; smaller = one leaf splits into pieces.
`leaf_cluster_min_points`	25	`--leaf_cluster_min_points`	DBSCAN min_samples (non-adaptive) and minimum cluster size to be kept.	Higher = fewer small fragments, small leaves may be lost.
`leaf_max_clusters`	50	-	Keep at most the 50 largest clusters (the colour palette has 50 colours).	Rarely needs changing.
`use_adaptive_dbscan`	True	`--no_adaptive_dbscan (turns off)`	Derive eps / min_samples from this cloud's point spacing.	Off = use the fixed values above.
`dbscan_eps_nn_factor`	2.5	`--dbscan_eps_nn_factor`	Adaptive eps candidate = factor x median nearest-neighbour distance.	Higher = larger eps, more merging.
`dbscan_eps_voxel_factor`	4.0	`--dbscan_eps_voxel_factor`	Adaptive eps candidate = factor x voxel_size (16 mm). The larger candidate is used.	Higher = larger eps, more merging.
`split_nearby_leaves`	True	`--no_split_nearby_leaves (turns off)`	Split clusters that contain several touching leaves, using their angle around the stem.	Off if single leaves are being wrongly split.
`leaf_split_min_cluster_size`	3000	`--leaf_split_min_cluster_size`	Only clusters with at least this many points are considered for splitting.	Lower = more clusters checked for splitting.
`leaf_split_min_angle_deg`	40	`--leaf_split_min_angle_deg`	Only split if the cluster spans at least 40 deg around the stem.	Lower = splits more easily; higher = only obvious merges are split.
`leaf_split_max_k`	3	-	Try splitting a cluster into at most 3 parts (KMeans k = 2..3).	Raise if more than 3 leaves are merged.

Step 7 - Leaf midrib

Midrib = shortest path through the leaf graph from the stem junction to the farthest point (the leaf tip).

Parameter	Default	Command-line flag	What it does	Effect of changing it
`leaf_graph_k`	18	`--leaf_graph_k`	k of the per-leaf k-NN graph used to find the midrib.	Higher = straighter path; lower = follows surface.
`leaf_graph_max_edge`	0.075 m	`--leaf_graph_max_edge`	Edges longer than 7.5 cm are removed from the leaf graph.	Lower = path cannot jump across gaps (tips may be missed); higher = path may shortcut.

Step 8 - Skeleton smoothing

Savitzky-Golay smoothing of the stem centerline and the leaf midribs.

Parameter	Default	Command-line flag	What it does	Effect of changing it
`smooth_window`	7	-	Filter window length in points.	Higher = smoother skeleton, slightly shorter lengths.
`smooth_polyorder`	2	-	Polynomial order of the filter (2 = local quadratic fit).	Higher = follows bends more closely.

Step 10 - Leaf width and area

The leaf is cut into slices along its midrib. Width of a slice = side-to-side spread of its points across the midrib. Leaf width = widest slice; leaf area = sum of slice width x slice length.

Parameter	Default	Command-line flag	What it does	Effect of changing it
`leaf_width_slice`	0.01 m	-	Slice length along the midrib (1 cm).	Smaller = finer width profile but noisier; larger = smoother.
`leaf_width_min_slice_points`	5	-	Slices with fewer points are skipped (width 0).	Lower for sparse leaves.
`leaf_width_percentile`	2.0	-	Width = spread between the 2nd and 98th percentile of the side offsets (ignores stray points).	Lower = closer to full edge-to-edge width but sensitive to noise.

Run options (command line only)

These control input, output and which parts of the pipeline run.

Flag	Default	What it does
`--bin`	P147/.../LtR_12th_Plant_230621pc_kinematic_calibration.bin	Ground-truth plant (.bin). Its points are also the input geometry. A bare file name is looked up in P147/manual_segmented_leaf_file/.
`--input`	None	Only with --skeleton_only: run on a .las / .ply / .pcd file instead of the .bin.
`--output_dir`	<bin folder>/iou_evaluation/<bin name>	Where results (CSV, PLY, GEXF) are written.
`--min_match_iou`	0.05	Predicted / ground-truth leaf pairs with IoU below this count as unmatched.
`--skeleton_only`	off	Run only the structure part (no classification / IoU).
`--classification_only`	off	Run only the classification + IoU part (no structure, no leaf area).
`--show_o3d_steps`	off	Open an Open3D window after each structure step.
`--no_viewer`	off	Do not open the PyVista viewer windows.

Where to start when results look wrong

Symptom	Parameters to try
Stem includes leaf bases	Raise stem_score_threshold, lower stem_radial_max, raise region_grow_normal_sim_min.
Stem is broken / too short	Lower stem_score_threshold, lower stem_seed_linearity_min, keep region growing on.
Two leaves merged into one	Lower leaf_cluster_eps / dbscan factors, lower leaf_split_min_angle_deg.
One leaf split into pieces	Raise leaf_cluster_eps / dbscan factors, or turn off split_nearby_leaves.
Small leaves missing	Lower min_leaf_component_size and leaf_cluster_min_points.
Leaf length too short / tip missed	Raise leaf_graph_max_edge.
Leaf width / area too small	Lower leaf_width_percentile, or use a smaller voxel_size.
Too noisy / too slow	Raise voxel_size, raise sor/ror strictness.